

Thesis Proposal

Topic: Deep Learning-Based Detection of Osteopenia and Osteoporosis from Knee X-Ray Images Using Hybrid CNN-Transformer Architectures

1. Introduction / Background

Osteopenia and osteoporosis are progressive skeletal disorders characterized by reduced bone mineral density and deterioration of bone microarchitecture. This research proposes a hybrid CNN-Transformer framework integrated with GAN-based augmentation and Explainable AI techniques for automated classification of knee X-ray images into normal, osteopenia, and osteoporosis categories.

2. Problem Statement

Conventional CNN architectures primarily focus on local feature extraction and often fail to effectively capture long-range spatial dependencies and global anatomical relationships within knee X-ray images. These limitations reduce sensitivity for early-stage disease detection and multiclass classification performance.

3. Objectives

1. Develop a hybrid CNN-Transformer framework for multiclass classification of knee X-ray images.
 2. Improve early-stage osteopenia detection using adaptive region-focused feature extraction.
 3. Enhance model interpretability using Grad-CAM and SHAP.
 4. Evaluate GAN-based augmentation for addressing class imbalance.
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4. Research Questions

1. Can hybrid CNN-Transformer architectures improve multiclass classification performance?
 2. How effectively can adaptive region-focused feature extraction capture trabecular and cortical bone variations?
 3. Does GAN-based augmentation improve sensitivity and generalization?
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5. Hypotheses

H1: The proposed hybrid CNN-Transformer framework combined with GAN-based augmentation will improve classification performance compared to conventional CNN models.

H2: Explainable AI methods such as Grad-CAM and SHAP will provide clinically meaningful visual explanations.

6. Research Methodology

The study adopts an experimental research design using publicly available knee X-ray datasets. Preprocessing techniques include image resizing, normalization, and augmentation. Models such as VGG19, ResNet, Inception, Xception, Vision Transformers, and hybrid CNN-Transformer architectures will be evaluated using accuracy, precision, recall, F1-score, sensitivity, specificity, and AUC-ROC metrics.

7. Significance of the Study

This research aims to provide a low-cost and accessible osteoporosis screening solution using routinely available knee X-ray images. The proposed framework may support radiologists in improving diagnostic accuracy and enabling early osteoporosis detection in resource-limited healthcare settings.

8. Ethical Considerations

Only de-identified retrospective imaging data will be used. Institutional approvals and ethical guidelines will be followed. The proposed system will serve as a clinical decision support tool rather than replacing medical professionals.
